

Multivariable Calculus Final Project

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Documentation

Introduction & Design

For this project, our group chose a cone with radius 30" and height 36" (or 2.5' and 3', respectively). The equation that we used was

$$\frac{x^2}{900} + \frac{y^2}{900} = \frac{z^2}{1296},$$

where $x \in [-30, 30]$, $y \in [-30, 30]$, and $z \in [0, 36]$.

Calculations

Theoretical Precise Value

To calculate the precise theoretic value, we will use the triple integral

$$\iiint_Q f(x, y, z) dV.$$

To find the bounds, we'll use cylindrical coordinates, as Cartesian coordinates can get messy. To find the bounds in cylindrical coordinates, we only need to solve for r , as $\theta \in [0, 2\pi]$ and $z \in [0, 36]$, as given by our domain restrictions above.

To find the bounds for r , we first convert the rectangular x and y to cylindrical coordinates, returning r . Thus, the cone equation becomes

$$\frac{r^2}{900} = \frac{z^2}{1296}.$$

Solving for r :

$$r = \sqrt{\frac{900z^2}{1296}} = \sqrt{\frac{25z^2}{36}} = \frac{5z}{6}.$$

Thus, the upper bound for r is $\frac{5z}{6}$, and because the lower bound for r is 0, we know that the final bounds for r are $r \in [0, \frac{5z}{6}]$. Thus, the complete bounds are:

$$\theta \in [0, 2\pi], \quad z \in [0, 36], \quad \text{and} \quad r \in \left[0, \frac{5z}{6}\right].$$

Therefore, the volume integral for the cone is If we plug in all the values for the triple integral, we end up with

$$\int_0^{2\pi} \int_0^{36} \int_0^{\frac{5z}{6}} r \, dr \, dz \, d\theta,$$

which we can solve as follows:

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^{36} \int_0^{\frac{5z}{6}} r \, dr \, dz \, d\theta = \int_0^{2\pi} \int_0^{36} \left[\frac{r^2}{2} \right]_0^{\frac{5z}{6}} dz \, d\theta = \int_0^{2\pi} \int_0^{36} \frac{25z^2}{72} dz \, d\theta \\ &= \int_0^{2\pi} \left[\frac{25z^3}{216} \right]_0^{36} d\theta = 5040 \int_0^{2\pi} d\theta = 5040 (2\pi) = 10800 (2\pi) \approx \boxed{33929 \text{ inch}^3}. \end{aligned}$$